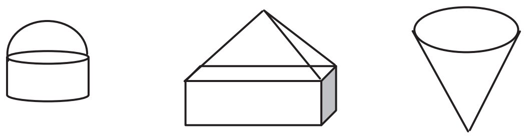
Higher

Ref	Content descriptor	Concepts and skills
GM z	Find circumferences and areas of circles	• Find circumferences of circles and areas enclosed by circles • Recall and use the formulae for the circumference of a circle and the area enclosed by a circle • Use $\pi \approx 3.142$ or use the π button on a calculator • Find the perimeters and areas of semicircles and quarter circles • **Calculate the lengths of arcs and the areas of sectors of circles** • **Answers in terms of π may be required** • Find the surface area of a cylinder
GM aa	Calculate volumes of right prisms and shapes made from cubes and cuboids	• Calculate volumes of right prisms, including the triangular prism, and shapes made from cubes and cuboids • Recall and use the formula for the volume of a cuboid • Find the volume of a cylinder • Use volume to solve problems
GM bb	**Solve mensuration problems involving more complex shapes and solids**	• **Solve problems involving more complex shapes and solids, including segments of circles and frustums of cones** • **Find the surface area and volumes of compound solids constructed from cubes, cuboids, cones, pyramids, spheres, hemispheres, cylinders** ***Examples:***  • **Solve problems including examples of solids in everyday use** • **Find the area of a segment of a circle given the radius and length of the chord**

Ref	Content descriptor	Concepts and skills
GM cc	**Use vectors to solve problems**	• **Understand and use vector notation** • **Calculate, and represent graphically, the sum of two vectors, the difference of two vectors and a scalar multiple of a vector** • **Calculate the resultant of two vectors** • **Solve geometrical problems in 2-D using vector methods** • **Apply vector methods for simple geometrical proofs**

4 Measures

What students need to learn:

Ref	Content descriptor	Concepts and skills
GM m	Use and interpret maps and scale drawings	• Use and interpret maps and scale drawings • Read and construct scale drawings • Draw lines and shapes to scale • Estimate lengths using a scale diagram
GM n	Understand **and use** the effect of enlargement for perimeter, area and volume of shapes and solids	• Understand the effect of enlargement for perimeter, area and volume of shapes and solids • Understand that enlargement does not have the same effect on area and volume • Use simple examples of the relationship between enlargement and areas and volumes of simple shapes and solids • **Use the effect of enlargement on areas and volumes of shapes and solids** • **Know the relationships between linear, area and volume scale factors of mathematically similar shapes and solids**
GM o	Interpret scales on a range of measuring instruments and recognise the inaccuracy of measurements	• Know that measurements using real numbers depend upon the choice of unit • Recognise that measurements given to the nearest whole unit may be inaccurate by up to one half in either direction

Ref	Content descriptor	Concepts and skills
GM p	Convert measurements from one unit to another	• Convert between units of measure in the same system (NB: Conversion between imperial units will be given. Metric equivalents should be known) • Know rough metric equivalents of pounds, feet, miles, pints and gallons: **Metric** — **Imperial** 1 kg — 2.2 pounds 1 *l* — $1\frac{3}{4}$ pints 4.5 *l* — 1 gallon 8 km — 5 miles 30 cm — 1 foot • Convert between imperial and metric measures • Convert between metric area measures • Convert between metric volume measures • Convert between metric speed measures • Convert between metric units of volume and units of capacity measures, eg 1 m*l* = 1 cm^3
GM q	Make sensible estimates of a range of measures	• Make sensible estimates of a range of measures in everyday settings • Choose appropriate units for estimating or carrying out measurements
GM r	Understand and use bearings	• Use three-figure bearings to specify direction • Mark on a diagram the position of the point B given its bearing from point A • Measure or draw a bearing between the points on a map or scaled plan • Given the bearing of a point A from point B, work out the bearing of B from A
GM s	Understand and use compound measures	• Understand and use compound measures, including speed and **density**
GM t	Measure and draw lines and angles	• Measure and draw lines, to the nearest mm • Measure and draw angles, to the nearest degree

Higher

Ref	Content descriptor	Concepts and skills
GM u	Draw triangles and other 2-D shapes using ruler and protractor	• Make accurate drawing of triangles and other 2-D shapes using a ruler and a protractor • Make an accurate scale drawing from a diagram • Use accurate drawing to solve a bearings problem